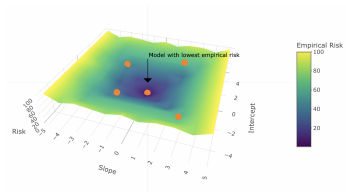
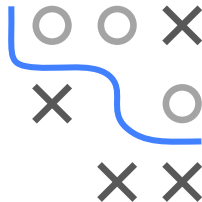


Introduction to Machine Learning

ML-Basics In a Nutshell



Learning goals

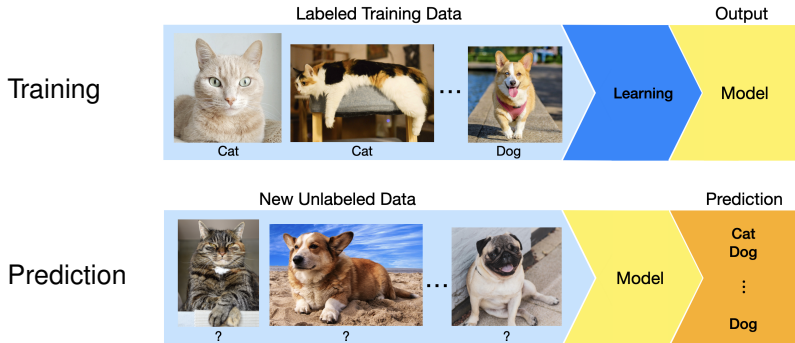
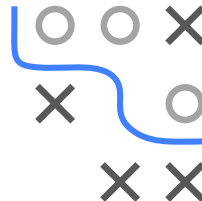
- Understand fundamental goal of supervised machine learning
- Know concepts of data, task, model, parameter, learner, loss function, and empirical risk minimization

WHAT IS ML?

“A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

Tom Mitchell, Carnegie Mellon University, 1998

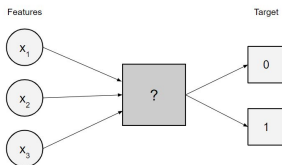
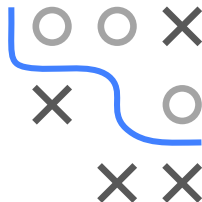
⇒ 99 % of this lecture is about **supervised learning**:



DATA IN SL

Measurements on different aspects of n objects:

- **Target:** output variable / goal of prediction
- **Features:** properties that describe an object



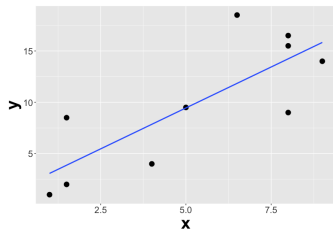
Features x				Target y
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.3	3.0	1.1	0.1	setosa
5.0	3.3	1.4	0.2	setosa
7.7	3.8	6.7	2.2	virginica
5.5	2.5	4.0	1.3	versicolor

$$\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)} \right), \dots, \left(\mathbf{x}^{(n)}, y^{(n)} \right) \right) \in (\mathcal{X} \times \mathcal{Y})^n$$

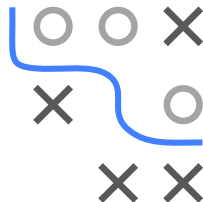
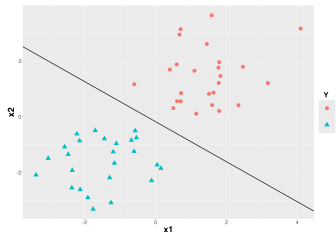
TASKS

- Supervised tasks are labeled data situations with the goal of learning the functional relationship between features and target
- We distinguish between **regression** and **classification** tasks, depending on whether the target is **numerical** or **categorical**

Regression: Target is **numerical**, e.g., predict days a patient has to stay in hospital



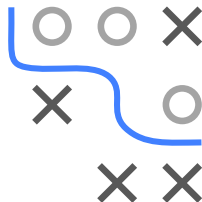
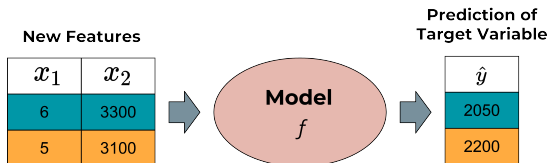
Classification: Target is **categorical**, e.g., predict one of two risk categories for a life insurance customer



MODELS AND PARAMETERS

- Maps features to predicted targets

$$f_{\theta} : \mathcal{X} \rightarrow \mathbb{R}^g$$

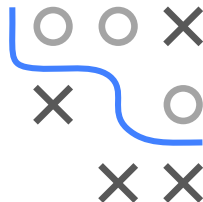


- We consider a restricted set of parametrized functions of a certain form, the **hypothesis space**, e.g., simple linear functions
- Models are fully determined by parameters. E.g., in the case of linear functions, $y = \theta_0 + \theta_1 x$, the parameters θ_0 (intercept) and θ_1 (slope) determine the relationship between y and x
- Finding the optimal model means finding the optimal set of parameters

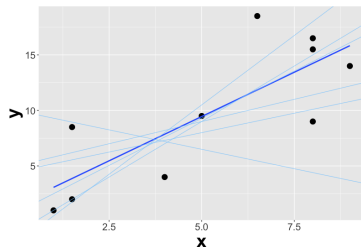
LEARNER

- Learns automatically the relation between features and target – given a set of training data
- Learner picks the best element of the **hypothesis space**, i.e., the function that fits the training data best:

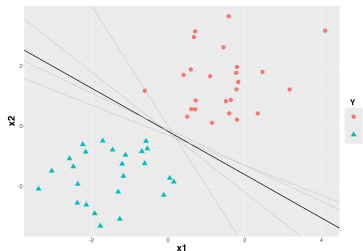
$$\mathcal{I} : \mathbb{D} \times \Lambda \rightarrow \mathcal{H}$$



Regression

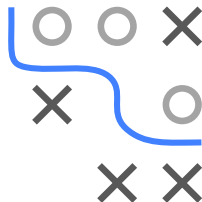


Classification



LEARNER

- Learner uses labeled training data to learn a model f . This model is applied to new unlabeled data for predicting the target variable



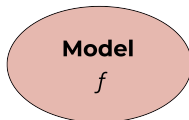
Train Set

y	x_1	x_2
2200	4	4300
1800	12	2700
1920	15	3100

Learner

New Features

x_1	x_2
6	3300
5	3100



Prediction of Target Variable



$\hat{y}$
2050
2200

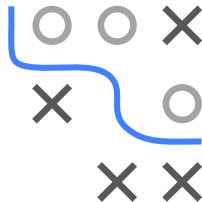
LOSS AND RISK MINIMIZATION

- Loss: Measured pointwise for each observation, e.g., L_2 -loss

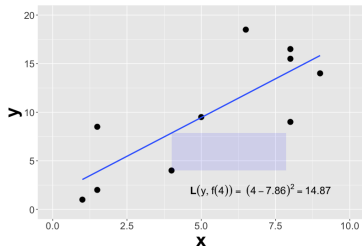
$$L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$$

- Risk: Measured for entire model. Sums up pointwise losses

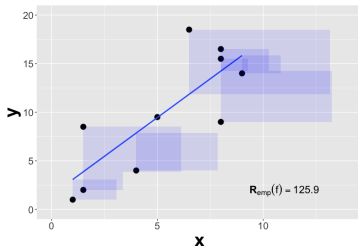
$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)}))$$



Squared **loss** of one **observation**



Empirical **risk** of entire **model**



EMPIRICAL RISK MINIMIZATION

- The risk surface visualizes the empirical risk for all possible parameter values of the parameter vector θ
- Minimizing the empirical risk is often done by numerical optimization

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta)$$

